

# Exam 1

● Graded

Student

Alejandro De La Torre

Total Points

27 / 50 pts

Question 1

1a

1 / 1 pt

✓ + 1 pt Correct

Question 2

1b

1 / 1 pt

✓ + 1 pt Correct or correct with respect to part (a)

Question 3

1c

1 / 1 pt

✓ + 1 pt Correct or correct with respect to part (a)

Question 4

1d

1 / 1 pt

✓ + 1 pt Correct or correct with respect to part (a)

Question 5

1e

1 / 1 pt

✓ + 1 pt Correct or correct with respect to part (a)

Question 6

2a

0 / 1 pt

✓ + 0 pts Incorrect or incomplete

Question 7

2b

0 / 1 pt

✓ + 0 pts Incorrect or incomplete

Question 8

2c

0 / 1 pt

✓ + 0 pts Incorrect or incomplete

Question 9

2d

1 / 1 pt

✓ + 1 pt Correct

Question 10

2e

1 / 1 pt

✓ + 1 pt Correct

Question 11

3

5 / 10 pts

✓ + 5 pts There is evidence of a reasonable basic proof outline, but there are multiple flaws in the argument.

- 1 The real numbers are closed under addition, but this does not mean that the irrational numbers are.
- 2 A proof by contradiction does not begin with an if...then statement. It begins with new hypotheses: the original hypotheses and the negation of the original conclusion.
- 3 Since we have proven in class that  $\sqrt{2}$  is irrational and the sum of a rational plus an irrational number is irrational.
- 4 This is not the correct negation of the given conclusion.

Question 12

4

5 / 10 pts

✓ + 5 pts There is evidence of a reasonable basic proof outline, but there are multiple flaws in the argument.

- 5 Base case is  $n=3$ .
- 6 For some.
- 7 You must prove this.

Question 13

5

3 / 10 pts

✓ + 3 pts An attempt was made, but proof has major errors or inconsistencies.

- 8 This is what you are trying to prove.
- 9 Why is this true?
- 10 The proof is incomplete.

Question 14

6

7 / 10 pts

✓ +7 pts Almost correct, but has a major flaw.

11 This is not exactly the distance you are looking for.

12  $N_1$  and  $N_2$  do not necessarily equal  $\epsilon/2$ .

## Exam 1

Name: Alejandro De La Torre

The exam contains 7 pages (including this cover page) and 6 questions. The total number of possible points is 50. You have 90 minutes to complete the exam.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **No outside resources allowed**. You may not use any notes, calculators, textbooks, or other resources on this exam. Make sure all electronic devices are silenced and put away.
- **Partial credit will be given**, so write down as much as you know even if some part of the proof might be missing.
- **Read the questions carefully**, make sure you answer the question being asked and understand the statement of the problem.
- **Answer all questions in the space provided**, you may use the back of pages as scrap paper, but only what is written under the question will be graded.

| Question | Points | Score |
|----------|--------|-------|
| 1        | 5      |       |
| 2        | 5      |       |
| 3        | 10     |       |
| 4        | 10     |       |
| 5        | 10     |       |
| 6        | 10     |       |
| Total:   | 50     |       |

Do not write in the table to the right.

1. (5 points) Consider the following subsets of  $\mathbb{R}$  and use them to answer the following questions. Write your answers in the boxes provided. **No justification is needed.**

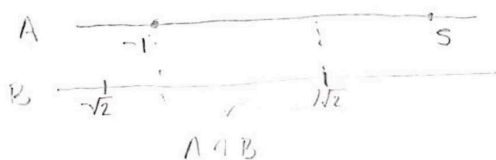
$$\bullet A = \{x \in \mathbb{R} : |x - 2| < 3\} \Rightarrow$$

$$\Rightarrow A = (-1, 5)$$

$$\bullet B = \{x \in \mathbb{R} : x^2 \leq 2\}$$

$$\Rightarrow [-\sqrt{2}, \sqrt{2}]$$

- (a) Express  $A \cap B$  as an interval in  $\mathbb{R}$ .



$$(-1, \sqrt{2}]$$

Find the following if they exist or indicate they do not exist:

- (b)  $\min(A \cap B)$

DNE

b/c -1 is not included

- lower bound can be -1 to  $-\infty$

- the greatest lower bound can be -1

$$DNE$$

- (c)  $\max(A \cap B)$

$$\sqrt{2}$$

- (d)  $\inf(A \cap B)$

greatest lower bound

$$-1$$

- (e)  $\sup(A \cap B)$

$$\sqrt{2}$$

2. (5 points) Determine if each of the following statements is TRUE or FALSE by completely filling in the corresponding bubble. No justification is needed.

(a) The negation of the statement " $x$  and  $y$  are both even integers" is " $x$  and  $y$  are both odd integers".

☒ TRUE

☐ FALSE

$x$  is even and  $y$  is odd

$x$  is odd and  $y$  is odd

if  $x$  &  $y$  are both not even integers  
then they must be both odd

(b) All unbounded sequences are divergent.

All convergent sequences  
are bounded

☐ TRUE

☒ FALSE

All divergent sequences  
are unbounded → not sure the other  
way around

(c) All divergent sequences are unbounded.

☒ TRUE

☐ FALSE

(d) If a set  $S$  has a maximum value, then  $\sup S \in S$ .

$$\forall x \in S, \max(S) \geq x$$

☒ TRUE

☐ FALSE

(e) The negation of the statement "for all real numbers  $a$ , there exists a real number  $b$  such that  $ab = 1$ " is the statement "there exists a real number  $a$  such that for all real numbers  $b$ ,  $ab \neq 1$ ".

☒ TRUE

☐ FALSE

For all real numbers  $a$ , there exists a real number  $b$  such that,  $ab \neq 1$   
there exists a real number  $a$  such that for all real numbers  $b$ ,  $ab = 1$

negation  
converse

maybe this directly negates it → no, if that  $a$  really exists such  
that for all real numbers  $b$ ,  $ab \neq 1$   
then that would disprove the original  
statement

3. (10 points) Use a **proof by contradiction** to prove the following statement. If you do any scratch work on this page you should clearly indicate what should be graded as your final proof. Your proof should be clearly laid out and your logic should be easy to follow.

Prove: If  $x \in \mathbb{R}$ , then  $x + \sqrt{2}$  is irrational or  $x - \sqrt{2}$  is irrational.

We argue by contradiction. Assume by way of contradiction

that  $x \in \mathbb{R}$  then  $x + \sqrt{2}$  is rational such that  $x + \sqrt{2} = \frac{p}{q}$

where  $p, q \in \mathbb{Z}$  and  $q \neq 0$ . Then

$$x + \sqrt{2} = \frac{p}{q} \Rightarrow x = \frac{p}{q} - \sqrt{2}$$

Then  $x$  is irrational as we know  $\sqrt{2}$  is irrational. Thus this is a contradiction. Since  $\mathbb{R}$  is closed under addition, it follows that if  $x \in \mathbb{R}$ , then  $x - \sqrt{2}$  too is irrational as it too, if we assume  $x$  is rational by way of contradiction, then  $x = \frac{p}{q} + \sqrt{2}$  would reveal  $x$  to be irrational, rendering the same contradiction.

Scratch Work

$$(x + \sqrt{2})(x + \sqrt{2})$$

$$x^2 + x\sqrt{2} + x\sqrt{2} + 2$$

$$\rightarrow x^2 + 2x\sqrt{2} + 2$$

$$\text{for } (x + \sqrt{2})^2 = \frac{p^2}{q^2}$$

I know in class Prof. Work meant when we can say  $\sqrt{2}$  is irrational unless explicitly asked to prove it.

Here, to say

$\frac{p}{q} - \sqrt{2}$  is irrational because  $\sqrt{2}$  is irrational is sorta a risk but I'll take it.

4. (10 points) Use a **proof by induction** to prove the following statement. If you do any scratch work on this page you should clearly indicate what should be graded as your final proof. Your proof should be clearly laid out and your logic should be easy to follow.

Prove: If  $n \in \mathbb{N}$  and  $n \geq 3$ , then  $2^n > n + 4$ .

We prove by induction. If  $n \in \mathbb{N}$  and  $n \geq 3$

Base case:  $P_n: 2^n > n + 4$

$$2^3 = 8 > 7$$

If  $n = 4$  then it follows that,

$$2^4 > (4) + 4$$

$$\Rightarrow 16 > 8$$

Inductive Step:

Assume by way of induction that  $2^n > n + 4$

For  $n \in \mathbb{N}$ ,  $n \geq 3$ , then

$$2^{(n+1)} > (n+1) + 4$$

$$\Rightarrow 2^n \cdot 2 > n + 4 + 1$$

(this is literally  $P_n: 2^n > n + 4$ )

Thus "since  $n+1$  was arbitrary" and  $\mathbb{N}$  is closed under multiplication and addition then if  $n \in \mathbb{N}$ ,  $n \geq 3$ , then  $2^n > n + 4$ .  $\square$

Scratch Work

$$P_n: 2^n > n + 4$$

$\Rightarrow$  we want to show that

for  $P_{n+1}$  that

$$2^{(n+1)} > (n+1) + 4$$

Following the NS peano axiom of the  $\mathbb{N}$  that forms the basis of induction.

$(n+1) \geq 3$  right for

$2^n > n + 4$ ? no no way - stick to what you know that is a given truth

So basically we want to show that  $n+1$  follows from  $n$  case

where we multiply RHS by 2 and add 1 on RHS



Since  $\varepsilon > 0$ , then  $\lim \sqrt{1 + \frac{1}{n}} = 1$

$$\forall \epsilon > 0 \exists N_1 \forall n > N_1 \Rightarrow |s_n - s| < \epsilon$$

6. (10 points) Prove the following statement, using the precise definition for the limit of a sequence. Do NOT use any of the algebraic limit theorems or any previous homework problems. If you do any scratch work on this page you should clearly indicate what should be graded as your final proof. Your proof should be clearly laid out and your logic should be easy to follow.

Prove: If  $\lim s_n = c$  and  $\lim t_n = c$ , then  $\lim |s_n - t_n| = 0$ . this is new.

Let  $\epsilon > 0$ . If  $\lim s_n = c$  then there exists  $N_1 = \frac{\epsilon}{2}$  such that  $n > N_1$  then

$$|s_n - c| < \frac{\epsilon}{2}$$

Notice that  $|s_n - (-c + c)| \leq |s_n - c| + |t_n - c| < \epsilon$ . Then if  $\lim t_n = c$  there exists  $N_2 = \frac{\epsilon}{2}$  such that  $n > N_2$  then

$$|t_n - c| < \frac{\epsilon}{2}$$

Let  $n > \max\{N_1, N_2\}$  then  $|s_n - c| + |t_n - c| < \frac{\epsilon}{2} + \frac{\epsilon}{2} < \epsilon$ . Since  $\epsilon > 0$  was arbitrary then  $\lim |s_n - t_n| = 0$ . □

(Scratch work)

$$\text{notice } |s_n - t_n - (c - c)| \leq |s_n - c| + |t_n - c| < \epsilon$$

by triangle inequality

so we want to find  $N_1$  and  $N_2$  such that  $n > \max\{N_1, N_2\}$

$$|s_n - t_n - c + c| \leq |s_n - c| + |t_n - c| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus since  $\epsilon > 0$  was arbitrary we can then conclude that  $\lim |s_n - t_n| = 0$

$$|s_n - c| + |t_n - c| < \epsilon = \frac{\epsilon}{2} + \frac{\epsilon}{2}$$

$\Rightarrow$

